

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1308

Claim Jumper 138 kV
December 15, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1308. The Interconnection Customer proposes a 133.6 MW battery energy storage facility interconnecting at the Claim Jumper 138 kV substation in the Pennsylvania area. The request was received on October 06, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 464 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$22,956,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
THERMAS – KILARC 138kV	TUSHAR – TEXAS EASTERN 5 345kV	162.9	177	5.1
MATANZAS – JOHN DAY 138kV	SIMPSON STUART TAP SUBSTATION – KEMPSTER 138kV Line	137.9	337	5.5
AUGUSTA ELECTRIC PLANT NO 1 – WINEAGLE 138kV Line	MEDORA – P.T.S 345kV L	122.8	349	5.6
TUCO SUBSTATION – JOHN DAY SUBSTATION 138kV	KEMPSTER – SIMPSON STUART TAP 138kV	122.0	389	5.5
CLAIM JUMPR – TUSHAR 138kV L	TUSHAR – DEMING 345kV L	117.8	211	64.4
SIMPSON STUART TAP – KEMPSTER 138kV Line	P.T.S – MEDORA 345kV	117.3	499	5.1
HOUGHTON ROCK – ENKA 138kV	ALLIGATOR 500/345 kV Transformer Bank	108.1	225	5.4
HOUGHTON ROCK – HALDDALE TAP 138kV	ALLIGATOR 500/345kV Auto	107.6	230	5.4
NELSON DEWEY SW YD 500-345 kV XFMR	TEXA EASTERN 5 – TUSHAR 345kV Line	104.9	612	13.9
EAST OMAK – STATION Q HARBOR 138kV	BURWELL – YOUNGWOOD 500kV	100.8	320	5.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
THERMAS – KILARC 138kV	Rebuild line with higher-capacity conductor	\$4,370,000	59% (\$2,582,000)
MATANZAS – JOHN DAY 138kV	Rebuild line with higher-capacity conductor	\$1,736,000	16% (\$274,000)
AUGUSTA ELECTRIC PLANT NO 1 – WINEAGLE 138kV	Rebuild line with higher-capacity conductor	\$4,190,000	29% (\$1,229,000)
JOHN DAY SUBSTATION – TUCO 138kV	Rebuild line with higher-capacity conductor	\$1,609,000	16% (\$254,000)
TUSHAR – CLAIM JUMPER 138kV	Rebuild line with higher-capacity conductor	\$14,507,000	60% (\$8,726,000)
SIMPSON STUART TAP – KEMPSEY SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$3,009,000	100% (\$3,009,000)
HOUGHTON ROCK – ENKA 138kV	Reconductor line	\$1,309,000	5% (\$71,000)
HOUGHTON ROCK – HALDALE TAP 138kV	Reconductor line	\$1,682,000	5% (\$91,000)
NELSON DEWEY SW YD 500 kV/345 kV Auto	Replace transformer with higher-rated unit	\$38,644,000	17% (\$6,556,000)
EAST OMAK – STA Q HARBOR 138kV	Reconductor line	\$2,158,000	8% (\$164,000)
	Total allocated to project		\$22,956,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.